Supplementary Material for *FedGAPS: Graph-Aligned Prototypes for Clustered Federated Learning under Domain Shift*

1 Overview

This appendix provides supplementary evidence and implementation details for FedGAPS. It is organized as follows. Section 2 reports the default hyperparameters used in the main experiments. Section 3 provides additional domain-level results. Section 4 examines the fuzzy collaboration structure across random seeds. Section 5 provides detailed prototype-geometry visualizations, and Section 6 complements the main-paper convergence analysis under milder label heterogeneity. Section 7 characterizes the recorded communication payload and training-loop time in the evaluated setting. Finally, Section 8 reports an additional collaboration-structure experiment and discusses the associated accuracy–communication trade-off. Unless stated otherwise, datasets, client partitions, optimization settings, and evaluation metrics follow the main paper.

2 FedGAPS Hyperparameters

Table 1 reports the default FedGAPS hyperparameters used in the main experiments. The sensitivity study in the main paper varies λ_g and β independently while holding the remaining parameters fixed.

3 Extended Predictive Results

3.1 Best- and Worst-Performing Domains

Table 2 complements the aggregate metrics in the main paper by identifying the domains that most frequently appear at the performance extremes. FedGAPS largely preserves the intrinsic difficulty ordering of the benchmarks rather than obtaining its gains from a favorable change in which domains are easiest or hardest. On Digits-5, MNIST remains the strongest domain and MNIST-M the weakest for FedGAPS and most baselines. On DomainNet-10, Real is consistently among the strongest domains, whereas Infograph is typically the weakest. On Office-10, DSLR frequently appears among the strongest domains, while Webcam and Caltech occur among the weaker domains. Together with the worst-domain results in the main paper, this shows that FedGAPS primarily raises performance on persistently difficult domains rather than changing the identity of the performance bottleneck. This distinction is important because an improvement in worst-domain accuracy is more informative when the underlying domain ordering remains broadly stable.

4 Soft Participation and Collaboration Structure

The fuzzy memberships in FedGAPS allow a client to retain nonzero association with more than one collaboration group. Figures 1–3 visualize this structure for $\alpha = 0.1$ across the three evaluation seeds. For seed 42, the aggregate soft shares are close to balanced on all three benchmarks: 50.3%/49.7% on Digits-5, 51.4%/48.6% on Office-10, and 52.4%/47.6% on DomainNet-10. By contrast, assigning each client only to its largest-membership cluster yields 60.0%/40.0%, 62.5%/37.5%, and 66.7%/33.3%, respectively. The seed-123 and seed-2024 visualizations show that the precise membership balance changes across runs, while non-negligible cross-cluster memberships remain visible.

This evidence is descriptive rather than a claim that balanced clusters are intrinsically preferable. Its main implication is that hard assignment removes information contained in secondary memberships. This interpretation is consistent with the main-paper ablation: replacing fuzzy memberships with hard assignments reduces mean accuracy in all six dataset–heterogeneity settings. The benefit of soft participation is therefore better understood as preserving overlapping client relationships for downstream training, rather than as enforcing equal cluster sizes.

Table 1: Default FedGAPS hyperparameters used in the main experiments.

Component	Parameter	Value
Reliability/support	$r_{\min}$	0.3
	τ_n	20
	β	0.5
Class balancing	γ	0.5
Discovery losses	λ_p	0.1
	λ_a	0.2
	λ_c	0.2
	λ_m	0.1
	λ_s	0.1
	λ_v	0.05
Graph/alignment	T	0.07
	T_g	0.2
	η	0.2
	λ_g	0.1
	ξ	10^{-4}
	μ	0.2
Client dissimilarity	α_{dist}	0.6
	λ_o	0.1
Discovery schedule	R_{ramp}	5
Model selection	W	1
	$\Delta_{\min}$	0

T denotes the contrastive temperature. α_{dist} is the prototype-distance mixing coefficient and is distinct from the Dirichlet concentration parameter $\alpha \in \{0.1, 0.5\}$.

5 Detailed Prototype Geometry

The main paper summarizes prototype geometry using mean same-class cross-domain and different-class within-domain cosine distances. Figures 4 and 5 provide the corresponding matrix-level views for representative settings. They distinguish three stages: the initial shared representation, the selected discovery checkpoint before graph smoothing, and the graph-smoothed representation at that checkpoint. For same-class cross-domain pairs, smaller distances indicate stronger cross-domain semantic alignment; for different-class within-domain pairs, larger distances indicate stronger class separation.

On DomainNet-10 with $\alpha = 0.5$, discovery reduces the same-class cross-domain distance and graph smoothing contracts it further, while the different-class distance remains above its initial level after smoothing. Thus, the graph primarily improves cross-domain consistency without collapsing the class structure learned during discovery. On Office-10 with $\alpha = 0.1$, the unsmoothed discovery representation initially increases the same-class distance, after which graph smoothing reduces it below the initial level. This case is particularly informative because the improvement cannot be attributed to discovery alone: the additional contraction appears after the graph-smoothing step. Overall, the matrix-level evidence supports the more cautious conclusion from the main paper: graph smoothing preferentially contracts same-class representations across domains, while its effect on inter-class separation depends on the dataset and heterogeneity level.

6 Additional Convergence Results

Figure 6 complements the $\alpha = 0.1$ convergence curves in the main paper with the milder $\alpha = 0.5$ setting. Across Digits-5, Office-10, and DomainNet-10, FedGAPS remains within the same overall communication-round horizon as the comparison methods. The figure is intended to characterize optimization behavior rather than to establish lower communication cost: FedGAPS includes a discovery phase and may exchange more than one cluster model for clients with retained memberships in multiple clusters.

7 Recorded Communication and Computation Cost

FedGAPS performs summary-based collaboration discovery after a shared representation-learning stage rather than through one-shot profiling. The $O(|\mathcal{Y}_i|p)$ term in the main paper describes only the per-client prototype-summary payload at each prototype exchange. Total training communication also includes model transfers during discovery and cluster-specific training, including multiple cluster-model exchanges when a client retains membership in more than one cluster.

Table 3 reports the quantities recorded by the implementation for $\alpha = 0.1$ and seed 42. Communication is accounting of uncompressed training tensors rather than measured network traffic, and elapsed time is training-loop wall-clock time rather than end-to-end deployment latency. Validation/test traffic, protocol overhead, and server-only processing time are not included. These measurements therefore characterize the evaluated implementation but should not be interpreted as a general systems benchmark or as evidence that compact prototype summaries reduce total communication.

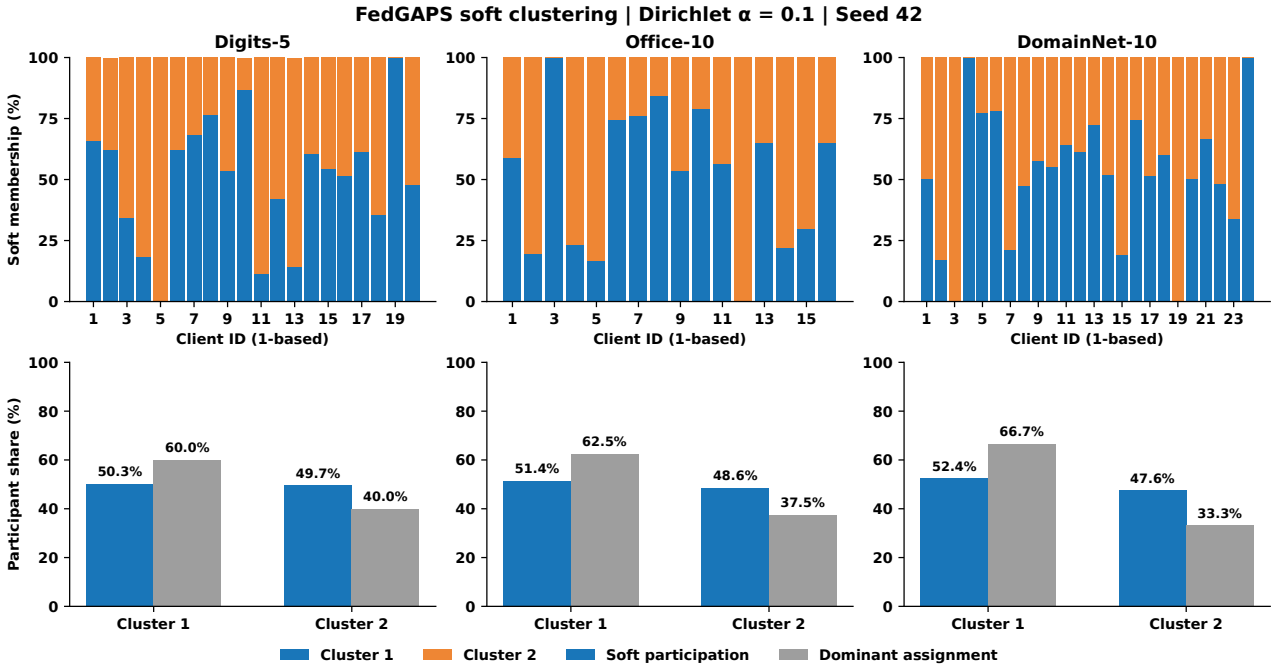


Figure 1: FedGAPS fuzzy memberships and aggregate participation for $\alpha = 0.1$, seed 42. Top: client membership weights, which sum to 100% for each client. Bottom: soft participation, $100N^{-1}\sum_i u_{ik}$, compared with the fraction of clients whose largest membership is assigned to cluster k. Soft participation is a fractional client share rather than a count of unique clients.

The current experiments involve 16–24 clients and 10 classes. Although the analytical server-side costs in the main paper depend on the number of clients, classes, domains, and representation dimension, the present measurements do not establish empirical scalability beyond this regime. We therefore restrict the cost claims to the evaluated setting.

8 Additional Collaboration-Structure Study

Table 4 provides an additional comparison of collaboration structures under the planted-group setting. This experiment is supplementary and is not used to support the headline results in the main paper. It compares histogram-based grouping, FedGAPS with $K \in \{2, 4\}$, domain-balanced FedAvg, and K-means applied to aligned signatures, while reporting both predictive performance and recorded communication.

On Digits-5, FedGAPS with $K = 4$ achieves the highest mean overall and worst-domain accuracy (82.93% and 56.32%). On DomainNet-10, FedGAPS with $K = 4$ and aligned signatures with K-means obtain similar overall accuracy (75.92% and 75.85%), with FedGAPS providing the highest mean worst-domain accuracy (37.62%). On Office-10, all methods are comparatively strong: FedGAPS with $K = 4$ gives the highest mean overall accuracy (91.58%), whereas domain-balanced FedAvg gives the highest mean worst-domain accuracy (88.93%). These results do not indicate that a larger K is universally preferable. Rather, they show that richer collaboration structures can improve predictive performance in some settings, while incurring additional model exchange and offering smaller gains when the benchmark is already well served by simpler collaboration. More broadly, the experiment illustrates the trade-off underlying collaborator selection: increasing the granularity of the collaboration structure can provide more specialized models, but the resulting predictive benefit must justify the additional training and communication cost. This complements the main-paper analysis, which fixes $K = 2$ to isolate the quality of the learned client relationships rather than tuning the number of groups separately for each dataset.

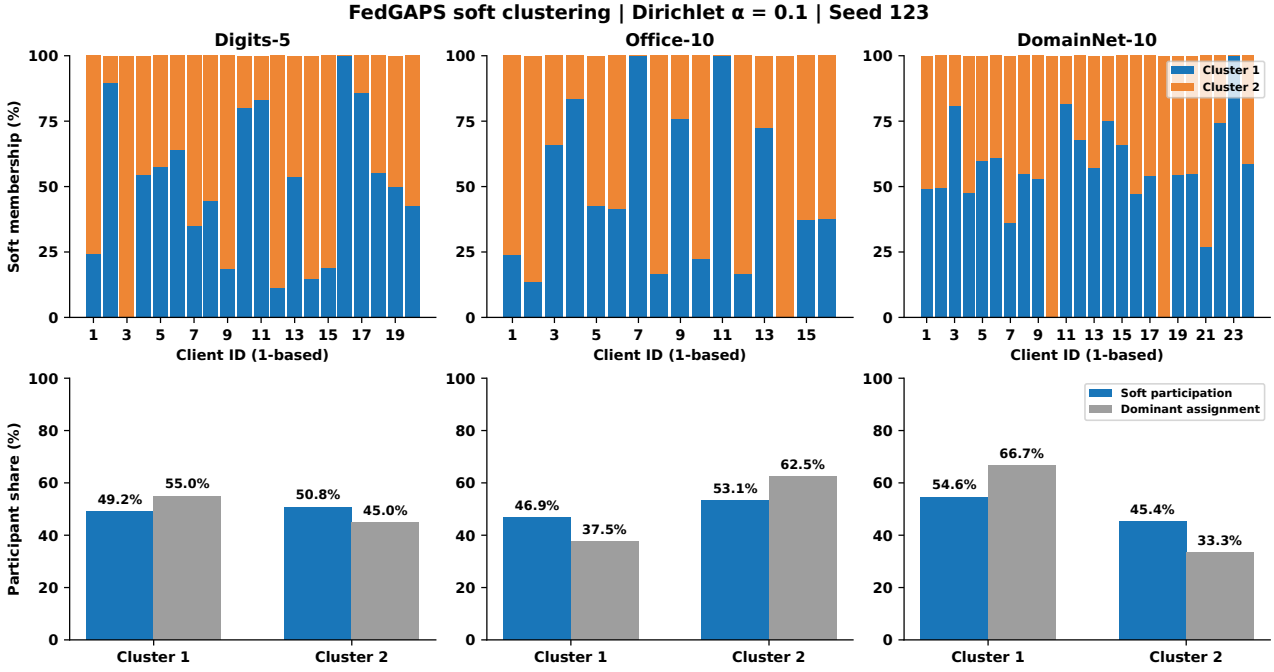


Figure 2: FedGAPS fuzzy memberships and aggregate participation for $\alpha = 0.1$, seed 123. The layout and participation definitions follow Fig. 1. Secondary memberships indicate clients retaining nonzero association with more than one collaboration group. Cluster identifiers are local to each run and do not imply matching cluster semantics across datasets or seeds.

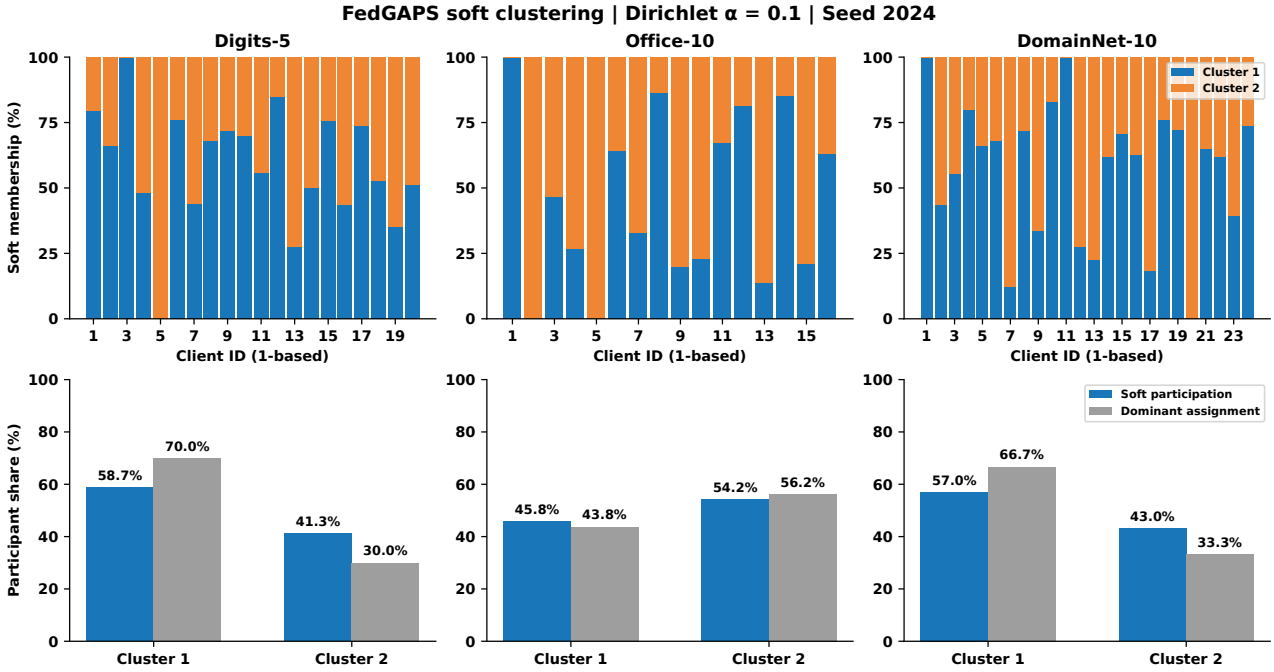


Figure 3: FedGAPS fuzzy memberships and aggregate participation for $\alpha = 0.1$, seed 2024. The layout and participation definitions follow Fig. 1. Secondary memberships indicate clients retaining nonzero association with more than one collaboration group. Cluster identifiers are local to each run and do not imply matching cluster semantics across datasets or seeds.

Table 2: Best- and worst-performing domains across datasets and heterogeneity settings. Multiple domains are reported when they occupy the corresponding extreme across the evaluated seeds.

	$\alpha = 0.1$		$\alpha = 0.5$	
Method	**Best domain(s)**	**Worst domain(s)**	**Best domain(s)**	**Worst domain(s)**
		Digits-5		
FedAvg	MNIST	MNIST-M	MNIST	MNIST-M
FedBN	MNIST	MNIST-M	MNIST	MNIST-M
FedProto	MNIST	MNIST-M, Synth	MNIST	MNIST-M
FPL	MNIST	MNIST-M	MNIST	MNIST-M
FedPLVM	MNIST	MNIST-M	MNIST	MNIST-M
FedDAP	MNIST	MNIST-M	MNIST	MNIST-M
FedRC	MNIST	MNIST-M	MNIST	MNIST-M
FedGAPS	MNIST	MNIST-M	MNIST	MNIST-M
		DomainNet-10		
FedAvg	Real	Info., Quick.	Real	Info.
FedBN	Real	Info.	Real	Info.
FedProto	Real	Info.	Real	Info.
FPL	Real	Info., Quick.	Real	Info.
FedPLVM	Clip., Real	Quick., Info.	Real	Info.
FedDAP	Real	Info., Quick.	Real	Info.
FedRC	Real	Info., Quick.	Real	Info.
FedGAPS	Real	Info.	Real	Info.
		Office-10		
FedAvg	Amazon, DSLR	Webcam, Amazon	DSLR	Webcam
FedBN	Amazon, Caltech	Webcam, DSLR	Amazon, DSLR	DSLR, Caltech
FedProto	DSLR	Caltech	DSLR	Caltech
FPL	Amazon, DSLR	Webcam, Amazon	DSLR	Webcam
FedPLVM	Amazon, DSLR	Webcam, Amazon	DSLR	Webcam
FedDAP	Amazon, DSLR	Webcam, Amazon	DSLR	Webcam
FedRC	Amazon, DSLR	Webcam	DSLR	Webcam
FedGAPS	DSLR	Webcam, Caltech	DSLR	Caltech, Webcam

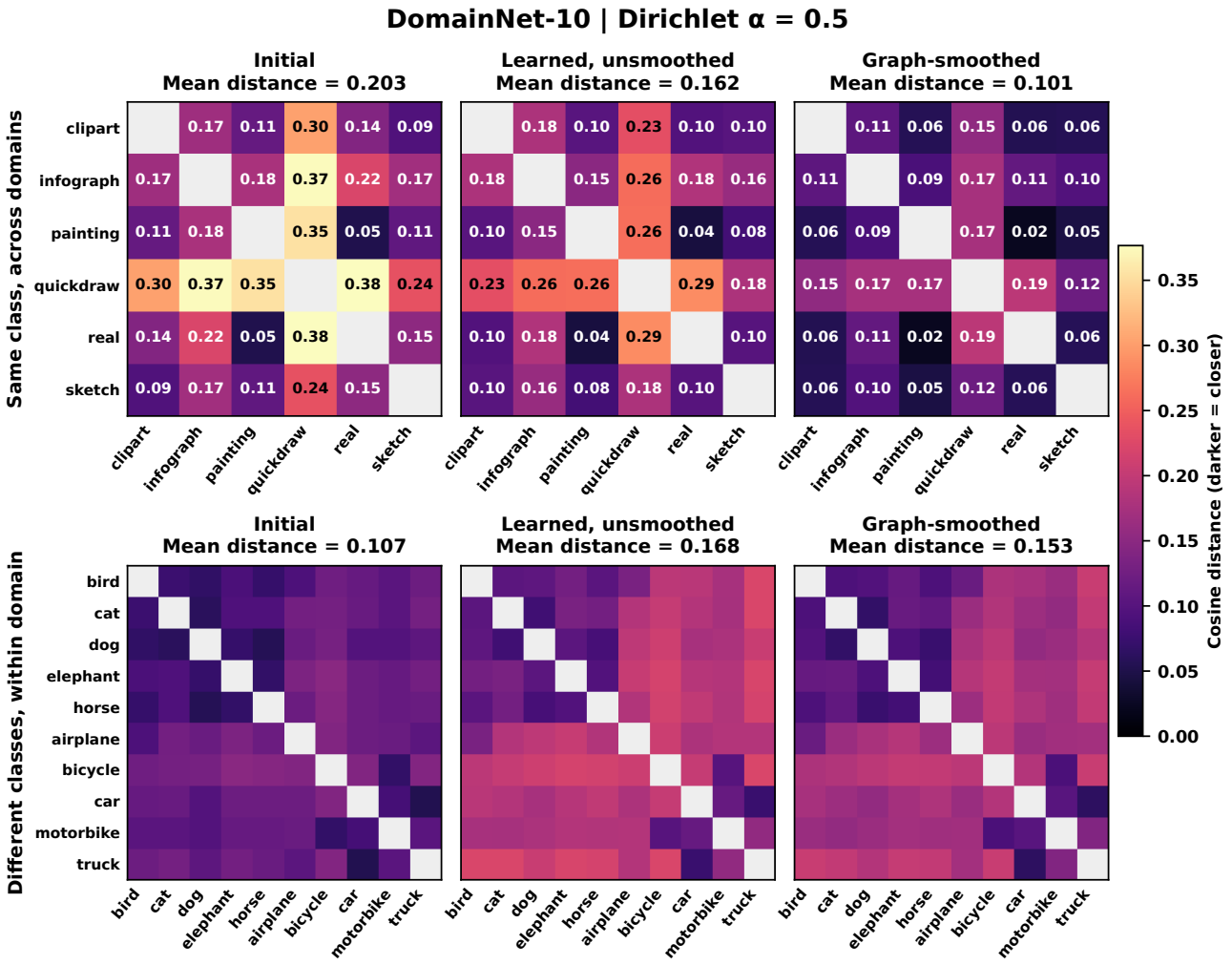


Figure 4: Prototype geometry on DomainNet-10 under Dirichlet $\alpha = 0.5$, averaged over three seeds. Columns show the initial representation, the selected discovery checkpoint before graph smoothing, and the graph-smoothed representation at that checkpoint. Top: same-class cross-domain cosine distances averaged over classes, where lower values indicate stronger cross-domain alignment. Bottom: different-class within-domain cosine distances averaged over domains, where higher values indicate stronger class separation. Diagonal entries are masked.

Table 3: Recorded FedGAPS training cost for $\alpha = 0.1$, seed 42. Communication counts uncompressed training-tensor payloads across discovery and cluster-specific training; GB denotes 10^9 bytes.

Dataset	Rounds	Time (s)	Communication (GB)		
			Upload	Download	Total
Digits-5	42	4096.8	48.53	67.41	115.94
Office-10	40	415.5	32.46	47.55	80.01
DomainNet-10	22	1739.7	33.13	41.76	74.89

Time denotes recorded training-loop elapsed time. Communication excludes validation/test traffic and transport overhead. Server-only processing time was not recorded.

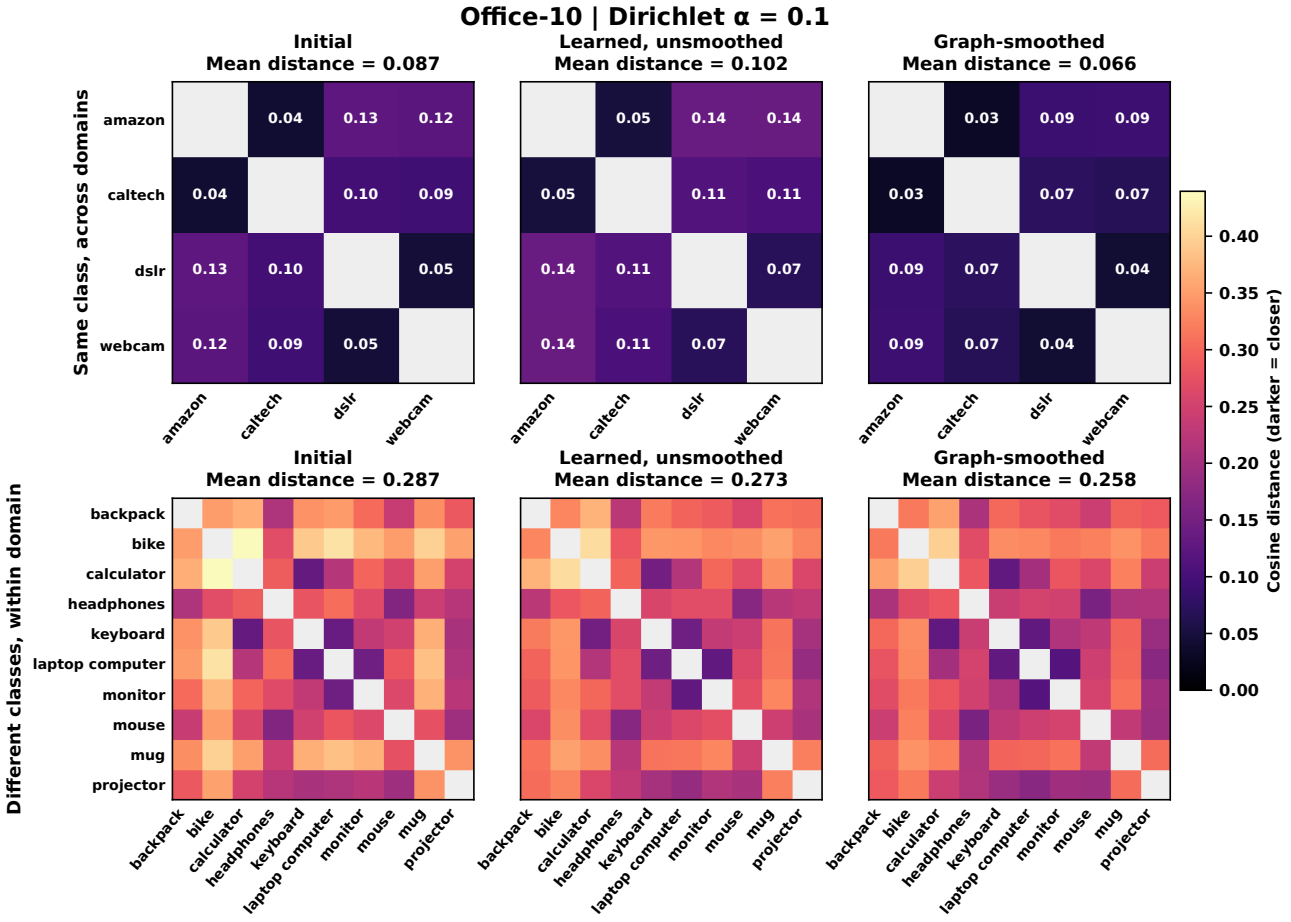


Figure 5: Prototype geometry on Office-10 under Dirichlet $\alpha = 0.1$, averaged over three seeds. Columns show the initial representation, the selected discovery checkpoint before graph smoothing, and the graph-smoothed representation at that checkpoint. Top: same-class cross-domain cosine distances averaged over classes, where lower values indicate stronger cross-domain alignment. Bottom: different-class within-domain cosine distances averaged over domains, where higher values indicate stronger class separation. Diagonal entries are masked.

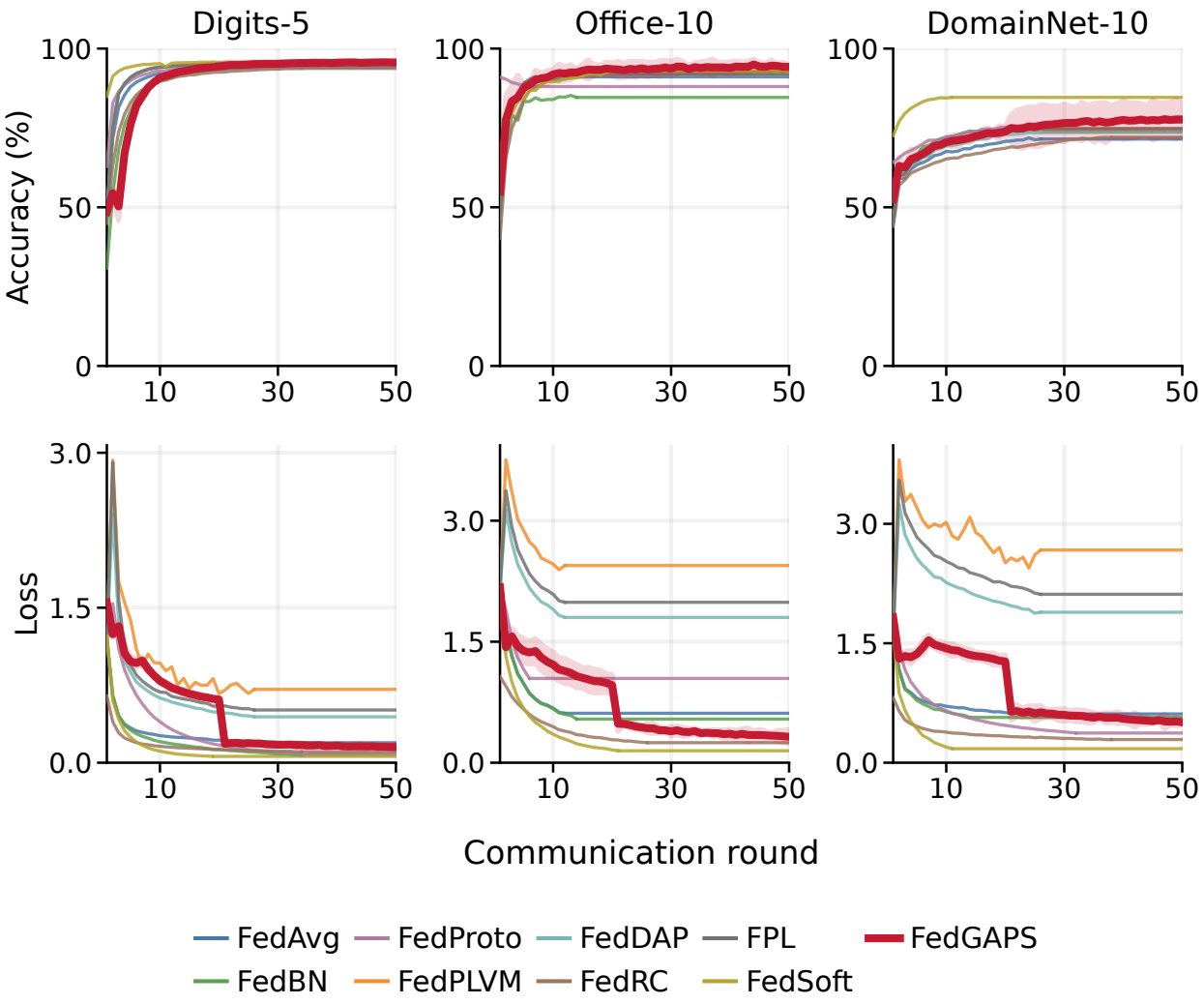


Figure 6: Accuracy and loss under Dirichlet $\alpha = 0.5$. Columns show Digits-5, Office-10, and DomainNet-10. The top row shows validation accuracy and the bottom row shows training loss. Curves average seeds 42, 123, and 2024 over their shared recorded rounds; horizontal extensions hold the last mean value to round 50 for visualization only. FedGAPS shading denotes two sample standard deviations.

Table 4: Predictive performance and recorded communication under the planted-group setting. Predictive metrics are mean $\pm$ sample SD over three seeds. Communication is uncompressed payload accounting.

Method	**Accuracy (%)**	**Worst domain (%)**	**Comm. (GB)**
	Digits-5		
Histogram-only, $K = 2$	74.36 ± 0.75	52.02 ± 0.79	108.87
FedGAPS, $K = 2$	77.02 ± 2.28	51.51 ± 1.87	147.37
Domain-balanced FedAvg	72.29 ± 1.26	49.15 ± 1.33	89.53
Aligned sig. + K-means, $K = 2$	75.79 ± 0.88	47.84 ± 2.64	108.43
FedGAPS, $K = 4$	82.93 ± 0.50	56.32 ± 0.02	150.06
	Office-10		
Histogram-only, $K = 2$	91.05 ± 0.19	83.52 ± 0.80	114.21
FedGAPS, $K = 2$	91.25 ± 0.28	86.36 ± 1.61	116.27
Domain-balanced FedAvg	91.51 ± 0.09	88.93 ± 1.19	71.62
Aligned sig. + K-means, $K = 2$	90.72 ± 1.21	85.23 ± 3.21	86.73
FedGAPS, $K = 4$	91.58 ± 1.12	87.70 ± 3.50	148.50
	DomainNet-10		
Histogram-only, $K = 2$	54.40 ± 1.64	27.67 ± 2.24	168.00
FedGAPS, $K = 2$	63.21 ± 10.22	29.91 ± 4.66	185.19
Domain-balanced FedAvg	62.95 ± 2.19	36.56 ± 5.31	107.44
Aligned sig. + K-means, $K = 2$	75.85 ± 0.82	36.96 ± 4.01	130.13
FedGAPS, $K = 4$	75.92 ± 0.69	37.62 ± 1.21	249.65